

Static Cosmology

Lecture 14: Automated Catalog Updating

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Overview

Catalog updates are fit cell by cell and accepted only when reference fields remain statistically consistent.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$r_{\text{cell}} = (r_{\text{color}}, r_{\text{flux}}, r_{\text{phase}}, r_{\text{shape}}, r_T)$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$\theta(x) = \theta_0 + \Delta\theta_c G_c(x)$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$\text{accept} = L_{\text{new}} \downarrow \wedge L_{\text{atlas}} \leq L_{\text{old}}$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Sky Cell Update Protocol
- ETHERMAP-33 loop guide
- CMT branch policy